

output:
pdf_document: default
html_document: default

ROI Framework v1 — Corporate Action AI (formula-first)

Status: framework for review. Numbers are placeholders to be replaced once the team and BNY confirm inputs. Companion workbook: `BNY_Corporate_Action_AI_ROI_Model_v1.xlsx`; longer notes in `roi-model-v1.md`. Formulas are written in LaTeX; a rendered PDF is attached alongside.

0. The question

Spend K now to build the system; does what it saves over 5 years justify K ?

$$\text{NPV} = \sum_{t=0}^5 \frac{\text{Net}_t}{(1+r)^t} \geq 0$$

Everything below defines Net_t .

1. Baseline: manual work that exists today (year $t = 1, \dots, 5$)

Symbol	Meaning
N_t	Event notifications in scope in year t ; N_1 = year-1 volume, g = annual growth
m	Share of events that are mandatory ($1 - m$ = voluntary)
s_M, s_V	Current STP rate for mandatory / voluntary events (share already fully automated)
τ_M, τ_V	Minutes of human handling per manually processed mandatory / voluntary event
h	Productive hours per FTE per year

c	Fully loaded annual cost per operations FTE
-----	---

$$N_t = N_1 (1 + g)^{t-1}$$

$$H_t = \frac{N_t}{60} \left[m (1 - s_M) \tau_M + (1 - m)(1 - s_V) \tau_V \right] \quad (\text{manual hours})$$

$$F_t = \frac{H_t}{h} \quad (\text{manual FTE}), \quad L_t = F_t \cdot c \quad (\text{addressable labour cost})$$

2. What the AI changes

Symbol	Meaning	Where the number comes from
a_t	Adoption share in year t (ramp from 0 to 1)	Rollout plan
α_M, α_V	Share of currently manual mandatory / voluntary events the AI fully automates	Our eval harness: share of events passing the confidence threshold and validated correct
β	Handling-time reduction on events that still need a human	Our eval / user test: review time with vs without AI pre-population
ρ	Realisation rate: share of released hours that becomes real savings or avoided hiring	BNY ops / finance

Manual hours remaining with the AI (non-adopted events keep full time; adopted events are either automated away or handled faster):

$$H'_t = \frac{N_t}{60} \left\{ m (1 - s_M) \tau_M [(1 - a_t) + a_t(1 - \alpha_M)(1 - \beta)] + (1 - m)(1 - s_V) \tau_V [(1 - a_t) + a_t(1 - \alpha_V)(1 - \beta)] \right\}$$

$$\Delta H_t = H_t - H'_t, \quad B_t^L = \frac{\Delta H_t}{h} \cdot c \cdot \rho \quad (\text{labour benefit})$$

Post-AI STP rate is derived, never assumed:

$$s'_{M,t} = 1 - (1 - s_M) (1 - a_t \alpha_M), \quad s'_{V,t} = 1 - (1 - s_V) (1 - a_t \alpha_V)$$

3. Error benefit

Symbol	Meaning
--------	---------

E	Baseline annual cost of corporate-action errors and losses in scope
δ	Share of that cost rooted in data sourcing / validation (the part the AI touches)
ε	Share of the data-rooted cost the AI eliminates at full adoption

$$B_t^E = E \cdot \delta \cdot \varepsilon \cdot a_t$$

4. Costs

Symbol	Meaning
K	One-time build and integration cost (year 0)
k	Annual run and maintenance, as a share of K
v	AI variable cost per event processed (LLM tokens, OCR, verification passes)
G	Annual governance, model-risk, platform and licence cost

$$C_0 = K, \quad C_t = K k + N_t a_t v + G \quad (t = 1, \dots, 5)$$

5. Outputs

$$\text{Net}_0 = -K, \quad \text{Net}_t = B_t^L + B_t^E - C_t$$

$$\text{NPV} = \sum_{t=0}^5 \frac{\text{Net}_t}{(1+r)^t}, \quad \text{DF}_t = \frac{1}{(1+r)^t}$$

$$\text{Payback} = \text{first } t \text{ with } \sum_{i=0}^t \text{Net}_i \geq 0 \quad (\text{interpolated within the year})$$

$$\text{IRR} = r^* \text{ such that } \sum_{t=0}^5 \frac{\text{Net}_t}{(1+r^*)^t} = 0$$

(Most Used) ROI on upfront investment — net operating benefit (after run costs) earned on the Year-0 build K ; cumulative over 5 years and annualised at steady state:

$$\text{ROI}_{\text{upfront}, 5y} = \frac{\sum_{t=1}^5 \text{Net}_t}{K}, \quad \text{ROI}_{\text{annual}, t} = \frac{\text{Net}_t}{K} \quad (\text{undiscounted})$$

(Optional) Benefit-cost ROI (Forrester TEI convention, kept only for comparison with vendor studies) — total cost in the denominator, including run costs:

$$\text{ROI}_{\text{TEI}} = \frac{\sum_{t=1}^5 (B_t^L + B_t^E) - \sum_{t=0}^5 C_t}{\sum_{t=0}^5 C_t} \quad (\text{undiscounted})$$

Break-even build cost — the most that could be spent on the build before NPV reaches zero (the run cost scales with K , hence the denominator):

$$K^* = K + \frac{\text{NPV}}{1 + k \sum_{t=1}^5 \text{DF}_t}$$

Payback and break-even build cost are the two numbers to use with a non-technical audience.

6. Finance terms used above (for non-finance readers)

FTE — full-time equivalent; two people at half time = 1 FTE. **Fully loaded cost** — salary plus benefits, tax, office, tooling; typically $1.4\text{--}1.6 \times$ base salary. **STP** — straight-through processing; an event handled end-to-end by systems with no human touch. **Discount rate** r — a dollar next year is worth less than a dollar today; r is the annual rate used to shrink future amounts (10% means each year's money counts roughly 10% less). **NPV** — net present value; all future net cash flows shrunk back to today's dollars, minus the upfront spend; positive means worth doing. **IRR** — internal rate of return; the annual return rate the project effectively earns, comparable with a company's hurdle rate. **Payback** — years until cumulative savings equal cumulative spend. **Realisation rate** — hours freed up are not automatically dollars saved; only the share that leads to not hiring or redeploying counts.